PATH FINDER

Line Follower Robot Competition

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EVENT BRIEF

Design, build, and program a fully autonomous robot to navigate a complex line-following track as fast and accurately as possible. Testing precision control, sensor engineering, and adaptive decision-making within a single 7-minute window.

VENUE & SCHEDULE

Venue Mechanical Science Block, 1st Floor | April 7, 2026

Time	Activity
10:00 – 10:15 AM	Event brief & participant verification
10:15 AM – 1:00 PM	Slot-wise competition
1:15 – 2:00 PM	Lunch
2:00 – 2:30 PM	Evaluation & extra slot (if needed)
3:30 PM	Certificate distribution

TEAMS

- 2 to 4 members per team
- Cross-department teams permitted; cross-college teams not allowed
- A participant cannot be part of more than one team
- Valid college ID mandatory on event day

ROBOT SPECIFICATIONS

Dimensions	Max 25 × 20 × 15 cm
Weight	Max 3 kg incl. battery
Power	Battery only, max 12V DC
Sensors	IR sensors only
Wireless	Bluetooth, Wi-Fi, RF prohibited

- Fully autonomous. Visible power switch mandatory.
- IR sensors must be shielded from the top. Semi-open venue, no rescheduling or extra time for ambient light interference.
- The robot must not scratch, mark, or damage the track.
- Kits, custom-built, and modified robots all permitted.

TRACK

- Black line, 38 mm ± 2 mm width, on a white surface
- Includes straight paths, sharp turns, intersections, loops, and dead ends

Track layout revealed at event start. No code or hardware changes permitted after reveal.

COMPETITION FORMAT

Each team gets 7 minutes on the track, make as many attempts needed within this window.

How does the 7-minute format work?
The track has intersections and dead ends; your robot won't know the correct path on the first attempt. That's intentional. Use early attempts to explore; later runs will be faster as the robot applies what it learned. The 7 minutes is your entire competition window; make it count.

SCORING

Official Score = **Best Run Time + Cumulative Penalties (all attempts)**

Penalty	Formula	Note
Line Crossing	(N ÷ 20) × Best Run Time	*N = times robot leaves line (all attempts)*
Restart	2n seconds	*n = manual repositions to start (all attempts)*
Adjustment	3e seconds	*e = manual motor/sensor adjustments (all attempts)*

- Best Run Time — fastest clean finish across all attempts
- Lowest official score wins. Tie-break: lower best run time wins.

Example: Best run = 40s. Left track 2×, restarted 3×, adjusted 2×:
40 + (2÷20)×40 + 2×3 + 3×2 = 40 + 4 + 6 + 6 = 56 seconds

DISQUALIFICATION

- Robot exceeds dimension or weight limits
- Any wireless or external control used during a run
- Code or hardware modified after track reveal
- Penalties accumulated to a degree the judge deems robot non-functional at judge's discretion.
- Robot damages the track

Judges' decisions are final. Event coordinators reserve the right to modify rules for smooth execution.
For queries, contact the organizing committee through contact@mecharush.in | mecharush.in